

A PROJECT REPORT

on

AI Text Detection

As a part of

Data Science [CE0630]

Submitted by

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BACHELOR OF TECHNOLOGY

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Problem Statement : AI Text Detection

Data Set : your_dataset_5000

1. Introduction

With the rapid growth of Artificial Intelligence, machines are now capable of generating human-like text. This creates challenges in identifying whether a piece of content is written by a human or generated by AI. The purpose of this project is to build a system that can distinguish between AI-generated and human-written text using machine learning and statistical analysis techniques.

2. Objective

- To analyze textual differences between AI and human writing
- To perform exploratory and statistical analysis on the dataset
- To build machine learning models for classification
- To evaluate model performance using multiple metrics
- To develop a prediction system for real-time text classification

3. Dataset Description

The dataset used in this project is named **your_dataset_5000**. It contains text samples labeled as:

- 0 → Human
- 1 → AI

Although the dataset consists of 5000 entries, a large number of records are duplicates. After removing duplicate entries and cleaning the data, the dataset reduces significantly to a smaller number of unique samples. This ensures that the model learns from meaningful and non-repetitive data.

4. Data Preprocessing

- Removal of duplicate rows
- Removal of missing or null values
- Conversion of text into lowercase
- Removal of special characters and unnecessary symbols

These steps standardize the text and make it suitable for machine learning models.

5. Feature Engineering

The features are described below:

- **Text Length:** Total number of characters in the text
- **Word Count:** Total number of words in the text
- **Average Word Length:** Measures writing complexity
- **Unique Word Ratio:** Indicates vocabulary diversity
- **Sentence Count:** Number of sentences in the text
- **Average Sentence Length:** Average number of words per sentence
- **Stopword Ratio:** Proportion of common words (e.g., “the”, “is”, “and”)
- **Capital Letter Ratio:** Proportion of uppercase letters
- **Punctuation Count:** Total number of punctuation marks
- **Exclamation Count:** Number of exclamation marks (!)
- **Question Count:** Number of question marks (?)
- **Digit Count:** Number of numeric characters
- **Long Word Count:** Number of words with more than 6 characters
- **Long Word Ratio:** Proportion of long words in the text

These features help capture important differences between AI and human writing. For example, AI-generated text often has more structured sentences, consistent word usage, and fewer emotional expressions, while human text may contain more variation, punctuation, and informal patterns.

6. Exploratory Data Analysis (EDA)

EDA is performed to understand the dataset visually:

- The dataset is balanced between AI and Human classes
- AI-generated text tends to be longer compared to human-written text

7. Feature Correlation Analysis

To better understand the relationships among the 14 engineered features, a correlation analysis was performed and visualized using a heatmap. This analysis helps in identifying how strongly different features are related to each other and whether any features provide similar or redundant information.

The correlation values range from **-1 to +1**, where:

- **+1** indicates a strong positive relationship (both features increase together)
- **-1** indicates a strong negative relationship (one increases while the other decreases)
- **0** indicates no relationship between the features

8. Statistical Analysis

Statistical techniques are applied to validate observations:

- **Skewness** is used to measure the asymmetry of data distribution
- **Kurtosis** is used to measure the presence of outliers
- **Z-score** is used to detect extreme values
- **Two-sample Z-test** is performed to check whether the difference between AI and human text is statistically significant

9. Outlier Detection and Removal

Outliers are detected using the Z-score method. Data points with extreme values are removed to improve model performance and reduce noise in the dataset. After removing outliers, the dataset becomes cleaner and more reliable for training.

10. Text Vectorization using TF-IDF

Machine learning models require numerical input, so text data is converted using TF-IDF :

- Captures importance of words
- Reduces influence of common words
- Uses unigrams and bigrams for better context understanding

This step transforms text into a numerical matrix suitable for model training.

11. Model Building

Three machine learning models are used in this project:

- Logistic Regression
- Naive Bayes
- Linear Support Vector Machine (SVM)

These models are chosen because they are efficient and perform well in text classification problems.

12. Model Evaluation

The models are evaluated using:

- Accuracy
- Cross-validation
- Precision, Recall, and F1-score
- Confusion Matrix
- ROC-AUC Curve

13. Feature Importance Analysis

The Naive Bayes model is used to identify important words:

- AI-related words include technical terms such as “data”, “models”, and “learning”
- Human-related words include conversational terms such as “my”, “me”, and “once”

This helps explain how the model differentiates between AI and human text.

14. Prediction System

A prediction function is implemented to:

- Take user input text
- Preprocess and vectorize it
- Predict whether the text is AI or Human
- Provide probability and confidence score

15. Key Findings

- AI-generated text is generally longer and more structured
- Human-written text is shorter and more natural
- Statistical analysis confirms clear differences between the two
- Machine learning models can accurately classify text

16. Limitations

- The dataset contains a large number of duplicate entries
- The number of unique samples is relatively small
- High accuracy may be due to limited data and repetition
- Model performance may decrease on unseen real-world data

17. Future Scope

- Use larger and more diverse datasets
- Apply deep learning models such as BERT
- Combine statistical and textual features for better performance
- Deploy the system as a web or mobile application
- Improve model generalization

18. Screen Shots of Project

```

*** (5000, 2)
Missing Values in: text      250
label      250
dtype: int64

Total missing values: 500

Total Duplicate alues in dataset: 4565

After Cleaning
Dataset Shape: (434, 2)

Label Distribution:
label
0.0    219
1.0    215
Name: count, dtype: int64

First 100 rows:

```

	text	label
0	Exercise plays a crucial role in supporting me...	1.0
1	Renewable energy helps fight climate change by...	1.0
2	A futuristic smart city is a vibrant, intercon...	1.0
3	Healthy eating habits are especially important...	1.0
4	Machine learning is transforming healthcare by...	1.0
...	...	...
95	LiDAR sensors provide precise three-dimensiona...	1.0
96	Chatbots integrated into educational apps offe...	1.0
97	Pilates strengthens the core muscles, improves...	1.0
98	Urban data platforms aggregate information fro...	1.0
99	Micronutrient deficiencies can impair cognitiv...	1.0

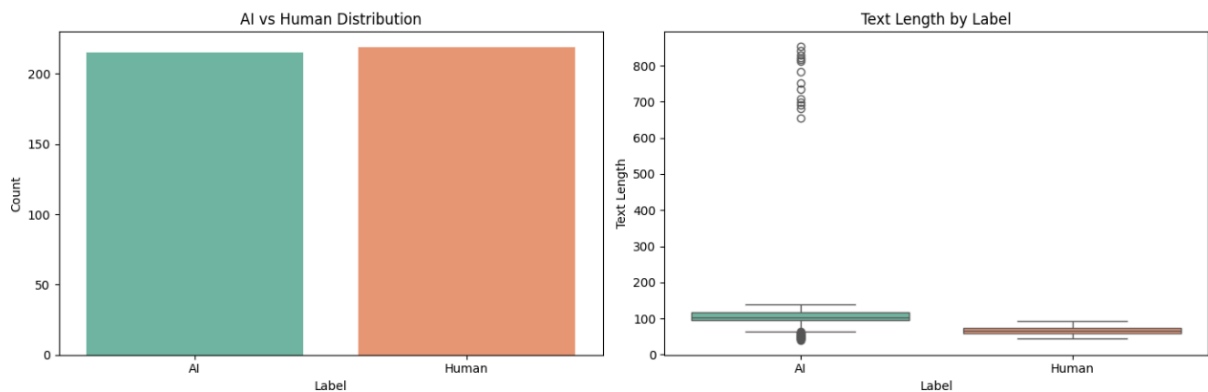
100 rows × 2 columns

Feature Statistics:

	text_length	word_count	avg_word_length	unique_word_ratio	sentence_count	avg_sentence_length	stopword_ratio
count	434.0000	434.0000	434.0000	434.0000	434.0000	434.0000	434.0000
mean	104.1912	15.4309	5.7466	0.9733	2.1382	6.4900	0.2923
std	123.1851	17.8883	1.2712	0.0542	0.7685	2.5288	0.1394
min	40.0000	5.0000	3.2500	0.6957	2.0000	2.5000	0.0000
25%	63.2500	11.0000	4.7000	0.9643	2.0000	5.5000	0.1778
50%	77.0000	13.0000	5.7247	1.0000	2.0000	6.5000	0.2857
75%	103.0000	14.0000	6.7644	1.0000	2.0000	7.0000	0.4000
max	853.0000	125.0000	10.1111	1.0000	7.0000	21.0000	0.6429

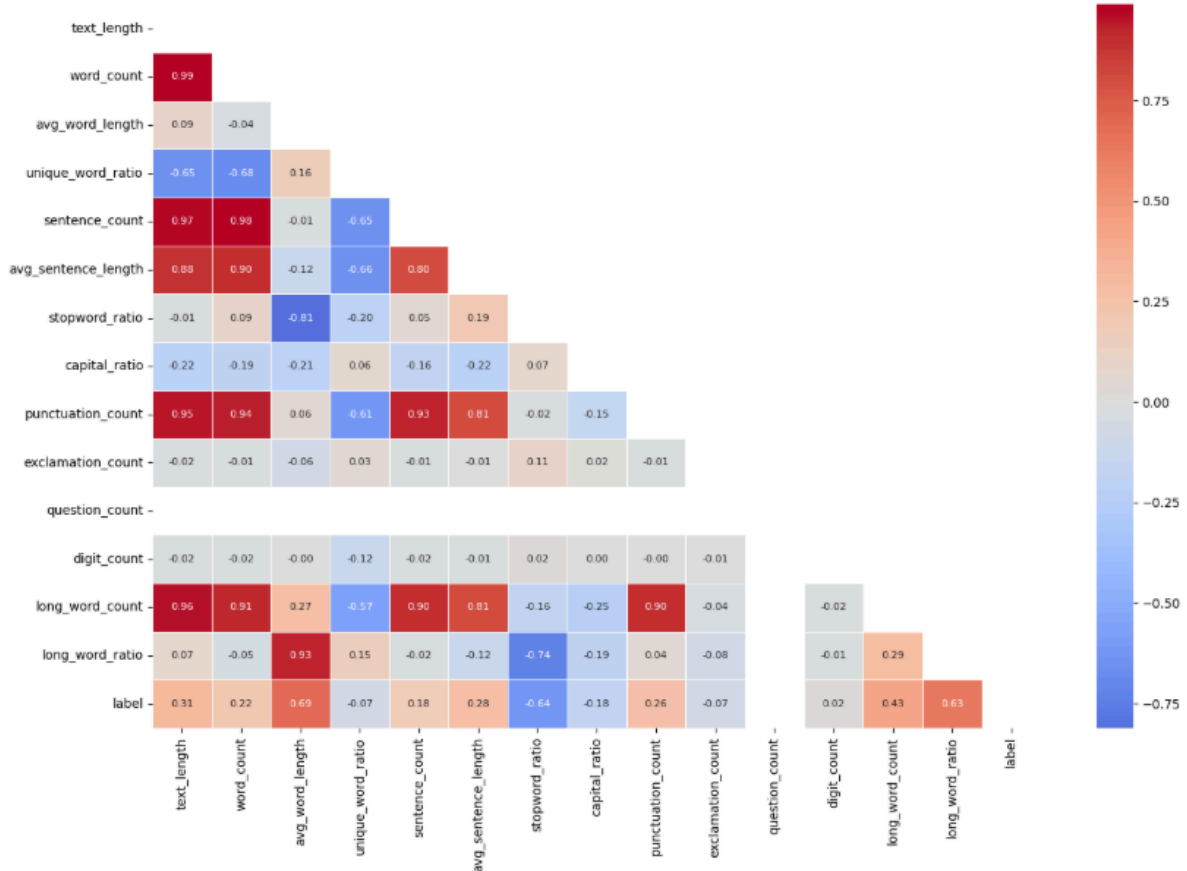
...

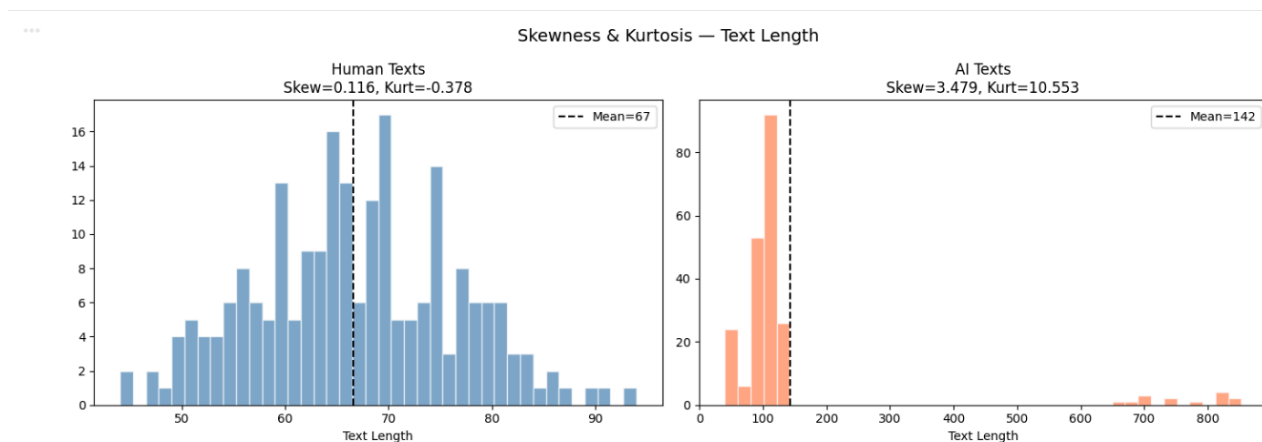
EDA — AI Text Detector



...

Feature Correlation Matrix





```

*** =====
Z-SCORE OUTLIER DETECTION (|z| > 3 = outlier)
=====
text_length      : 14 outliers (3.23%)
word_count       : 14 outliers (3.23%)
avg_word_length  : 1 outliers (0.23%)
unique_word_ratio : 15 outliers (3.46%)
sentence_count   : 14 outliers (3.23%)
avg_sentence_length : 14 outliers (3.23%)
stopword_ratio    : 0 outliers (0.00%)
capital_ratio     : 6 outliers (1.38%)
punctuation_count : 13 outliers (3.00%)
exclamation_count : 2 outliers (0.46%)
question_count    : 0 outliers (0.00%)
digit_count       : 4 outliers (0.92%)
long_word_count   : 12 outliers (2.76%)
long_word_ratio   : 2 outliers (0.46%)

Dataset after outlier removal: 420 rows
Rows removed: 14

```

```

*** =====
TWO-SAMPLE Z-TEST: AI vs Human (Key Features)
=====

text_length
  AI Mean : 99.2537
  Human Mean : 66.6073
  Z-Statistic: 19.6516 | P-Value: 0.000000
  Result : Significant (p < 0.05)

word_count
  AI Mean : 12.8955
  Human Mean : 11.5936
  Z-Statistic: 4.8341 | P-Value: 0.000001
  Result : Significant (p < 0.05)

unique_word_ratio
  AI Mean : 0.9827
  Human Mean : 0.9771
  Z-Statistic: 1.4101 | P-Value: 0.158523
  Result : Not Significant (p >= 0.05)

```

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--- Linear SVM ---
Test Accuracy : 0.9524
CV (5-fold) : 0.9429 ± 0.1025
precision    recall    f1-score    support

Human        0.98        0.93        0.95        44
AI           0.93        0.97        0.95        40

accuracy          0.95        84
macro avg         0.95        0.95        0.95        84
weighted avg      0.95        0.95        0.95        84

```

```

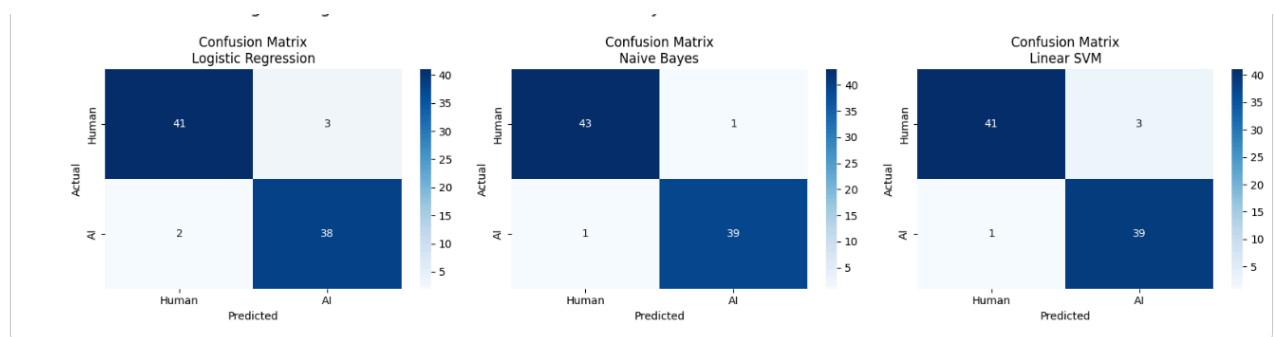
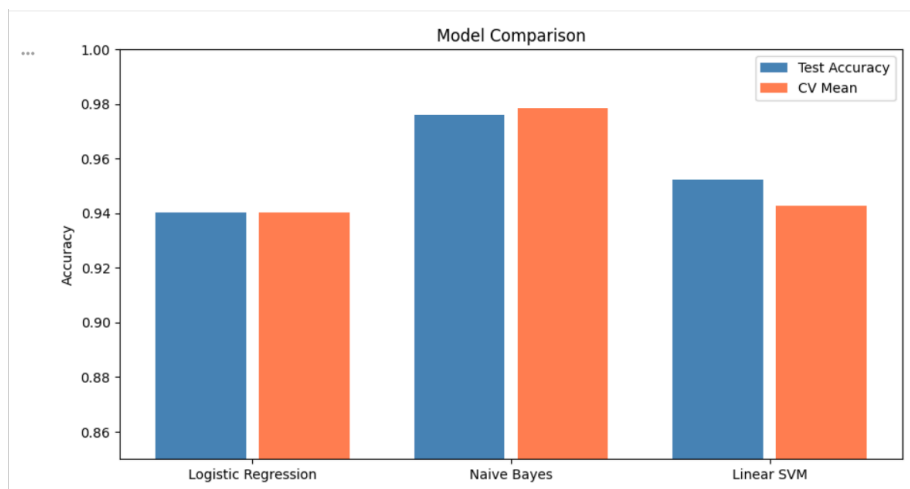
... --- Logistic Regression ---
Test Accuracy : 0.9405
CV (5-fold)   : 0.9405 ± 0.0955
              precision recall f1-score support
    Human      0.95      0.93      0.94      44
     AI        0.93      0.95      0.94      40

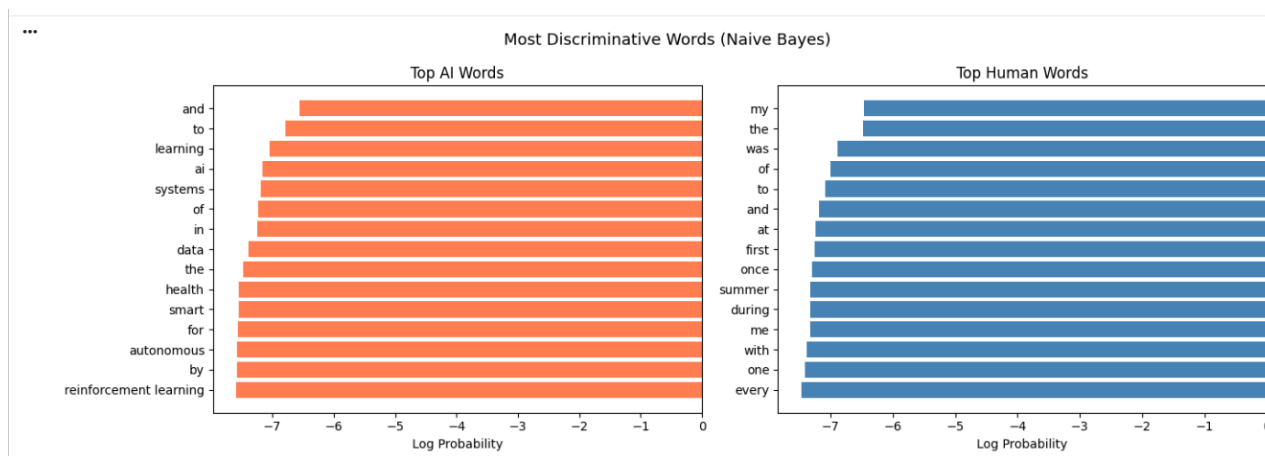
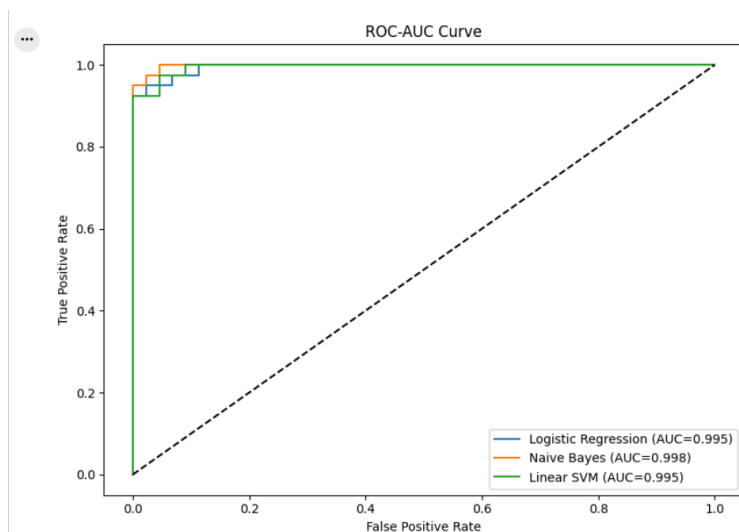
    accuracy
macro avg      0.94      0.94      0.94      84
weighted avg   0.94      0.94      0.94      84

--- Naive Bayes ---
Test Accuracy : 0.9762
CV (5-fold)   : 0.9786 ± 0.0314
              precision recall f1-score support
    Human      0.98      0.98      0.98      44
     AI        0.97      0.97      0.97      40

    accuracy
macro avg      0.98      0.98      0.98      84
weighted avg   0.98      0.98      0.98      84

```





```

... Enter text to detect:
Heyy Toady is the most Beautiful Morning I had ever seen. I like this place. What about you ??
===== RESULT =====
Label      : Human Written
AI %       : 33.3%
Confidence : 59.78%
(np.float64(0.0), 33.33333333333333, np.float64(0.5977896793957898))

```

19. Conclusion

This project successfully demonstrates how machine learning and statistical techniques can be used to detect AI-generated text. By analyzing writing patterns and using classification models, the system achieves high accuracy. Despite some dataset limitations, the project provides a strong foundation for further improvements and real-world applications.